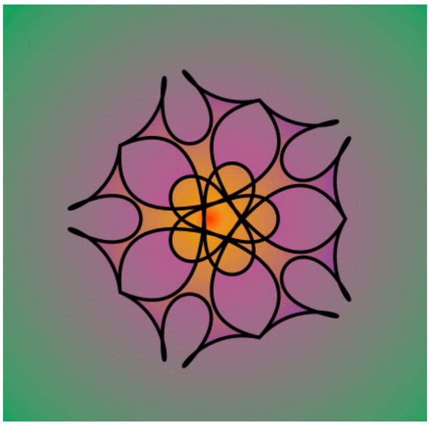

Group Theory

An Introduction to Algebraic Symmetry and Structures

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1 Introduction to Groups

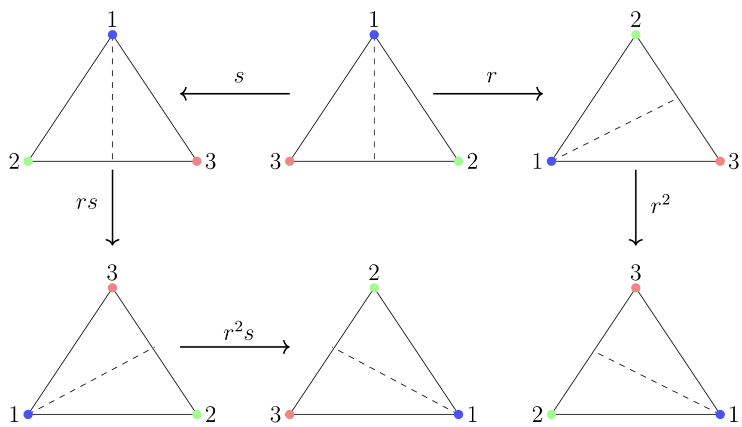
Definition 1.1 (Group). Let G be a set, and we define a binary operation $*$: $G \times G \to G$, then $(G, *)$ is said to be a group if it follows the following axioms:

1. $\forall\, a, b \in G \implies a * b \in G$, and we say the set has **closure**.
2. $\forall\, a, b, c \in G$ we have $a * (b * c) = (a * b) * c = a * b * c$, so the operation is **associative**.
3. $\exists\, e \in G \,\forall\, a \in G$ such that $a * e = e * a = a$, where e is called the **identity** of the group.
4. $\exists\, a^{-1} \in G \,\forall\, a \in G$ such that $a * a^{-1} = a^{-1} * a = e$, where a^{-1} is called the **inverse** of an element a.

That's a particularly heavy seeming definition, but it happens to be incredibly simple, and we show its simplicity by listing a number of examples:

Example 1.2.

1. Consider an equilateral triangle, lets label each corner with a colour. Now we perform operations on this triangle that change the order of the colours, which are *rotations* (r) and *reflections* (s).



   If we consider these *set* of operations to be elements, and take the binary operation to be performing one operation after another, then we have a group! This is generally called the dihedral group D_{2n}, where n is the number of sides the regular n–gon has. It is the group of symmetries that a regular n–gon possesses, n rotations and n reflections. The action of doing nothing is also an element of this group, as every group must have an identity.

2. The natural numbers $\mathbb{N}$, integers $\mathbb{Z}$, rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$ and complex number $\mathbb{C}$ are groups on the operation of addition.
3. On multiplication, only $\mathbb{Q}^*, \mathbb{R}^*$ and $\mathbb{C}^*$ are groups, since elements of $\mathbb{Z}^*$ do not have multiplicative inverses (reciprocals) lying in $\mathbb{Z}^*$. Here, $F^* = F - \{0\}$, since the inverse of zero is undefined in all sets.
4. Vector spaces on real or complex fields are also groups by defintion.
5. The set of $n \times n$ non–singular matrices with entries from a set X form a group upon matrix multiplication, and this is called the general linear group.

$$\mathrm{GL}(n, X) = \{A = (a_{ij})_{n\times n} \mid a_{ij} \in X, \det(A) \neq 0\} \tag{1}$$

6. All the valid twisting moves of a Rubik's Cube combined in sequence form a finite group.

7. Let us take the days of the week, and assign a number to each of them with Sunday being 0 and Saturday being 6. So we have a set $\{0, 1, 2, 3, 4, 5, 6\}$, now we define an operation where we add two days together, say $3 + 6 = 9$, but the day 6 days after Thursday is Wednesday, which is labelled by 2. So in general, the operation is addition modulo 7 (modulo 7 just means remainder after division by 7). We have an identity element 0, and the inverse for an element k would be $7 - k$. Addition is associative and the addition will will give us an element in the set. We have formed a group! In general this set, $\mathbb{Z}_n = \{0, 1, 2, \ldots n - 1\}$ is called the group of residue classes modulo n.

8. A similar group to $\mathbb{Z}_n$ is U_n, which contains elements which are coprime to n i.e. integers k with $\gcd(n, k) = 1$, with the operation being multiplication modulo n. For example: $U_{12} = \{1, 5, 7, 11\}$, with the inverse k^{-1} of an element k being the solution to $k^{-1}k = 1 \pmod{n}$.

9. The set of the complex solutions to $x^n = 1$, called the n^{th} roots of unity also form a group under complex multiplication.

$$\mathbb{C}_n = \left\{ e^{i2k\pi/n} \mid 0 \leq k < n \right\} \tag{2}$$

Now, we shall prove some basic lemmas of groups that we usually take for granted. From here on, we omit the $(*)$ when mentioning whether something is a group, and when taking the operation between elements.

Lemma 1.3.

1. **(Left Cancellation)** *$\forall\, a, b, c \in G$ such that $ab = ac \implies b = c$.*

2. **(Right Cancellation)** *$\forall\, a, b, c \in G$ such that $ba = ca \implies b = c$.*

3. *There is only one unique identity element for any group.*

4. *There is only one unique inverse for an element in a group.*

Proof.

1. Given that $ab = ac$, for some $a, b, c \in G$, by the existence of inverse, we write $a^{-1}(ab) = a^{-1}(ac)$ which turns into $a^{-1}ab = a^{-1}ac$ by associativity and into $eb = ec$ then we have $b = c$.

$\square$

2. Given that $ba = ca$, for some $a, b, c \in G$, by the existence of inverse, we write $(ba)a^{-1} = (ca)a^{-1}$ which turns into $baa^{-1} = caa^{-1}$ by associativity and into $be = ce$ then we have $b = c$.

$\square$

3. Let e' and e be two identities of a group G. Then, $ae = ea = a$ and $ae' = e'a = a \;\forall\, a \in G$. So we have $a = ae = ae'$ and by left cancellation we get $e = e'$, hence the two identities are one and the same.

$\square$

4. Let a' and a'' be two inverses of an element $a \in G$, then $aa' = a'a = e$ and $aa'' = a''a = e$. So we have $e = a'a = a''a$ and by right cancellation we have $a' = a''$, hence the two inverses are one and the same.

□

Definition 1.4 (Abelian Group). A group G is said to be an abelian group if and only if $\forall\, a, b \in G$, we have $ab = ba$, i.e. the operation is commutative on G. Hence, it is typically also called a commutative group. It was named after Niels Henrik Abel, one of the founders of group theory.

Definition 1.5 (Subgroup). Let G be a group, and we have a set $H \subset G$. Then, H is called a subgroup of G if and only if H is also a group, denoted by $H < G$.

Example 1.6.

1. Aside from the obvious ones like $\mathbb{Q} < \mathbb{R}$, lets look at the group $\mathrm{SL}(n, X)$, the special linear group which consists of non singular $n \times n$ matrices with unit determinant. This is a subgroup of the general linear group $\mathrm{GL}(n, X)$.

$$\mathrm{SL}(n, X) = \{A = (a_{ij})_{n\times n} \mid a_{ij} \in X, \det(A) = 1\} < \mathrm{GL}(n, X) \tag{3}$$

2. The group of integer multiples of an integer a as $\mathbb{Z}a$, which is a subgroup of $\mathbb{Z}$.

$$\mathbb{Z}a = \{n \in \mathbb{Z} \mid n = ka \text{ for some } k \in \mathbb{Z}\} \tag{4}$$

Proposition 1.7 (Subgroup Criteria).
i. *If G is group and $H \subset G$, then H is a subgroup if and only if $\forall\, a, b \in H \implies ab^{-1} \in H$.*

ii. *If G is a group and $H \subset G$, then H is a subgroup if and only if $\forall\, a, b \in H \implies ab \in H \quad$ and $a^{-1} \in H$.*

Proof.
i. ($\implies$) If H is a subgroup, then by closure and existence of inverse, $ab^{-1} \in H$ for every $a, b \in H$.
($\impliedby$) If $\forall\, a, b \in H \implies ab^{-1} \in H$, then $aa^{-1} = e \in H$, which means that $\forall\, a \in H$ we have $ea^{-1} = a^{-1} \in H$, and $\forall\, a, b^{-1} \in H$, we have $ab \in H$. This means we have existence of identity, inverse and closure, so H is group and since it is a subset of G, thus $H < G$.

□

ii. Both sides are directly implied by the group axioms. Thus, $H < G$. □

Definition 1.8 (Order). If we have a group G with a finite number of elements, then that number of elements is the order of the group, written as $|G|$. If the group is an infinite group, then $|G| = \infty$.
The order of an element $a \in G$ is the number of times it must be composed with itself to get the identity i.e. if $a^k = e$, then the order of a, written as $\mathcal{O}(a) = k$. If $a^k = e$ never occurs for any k, then $\mathcal{O}(a) = \infty$.

Lemma 1.9. *If G is a group, and we have $\mathcal{O}(a) = n$ for some $a \in G$, then if we have $a^m = e \implies n \mid m$ (n divides m).*

Proof. By the definition of order of an element, n is the smallest integer s.t. $a^n = e$. It follows that $(a^n)^k = a^{nk} = e$, and only multiples of n can give us an identity, Hence, m must also be a multiple of n and thus $n \mid m$. □

Lemma 1.10. *Let G be a group, and let $a \in G$ with $\mathcal{O}(a) = n$. Then $\mathcal{O}(a^k) = n/\gcd(n, k)$.*

Proof. Let $\mathcal{O}(a^k) = p$, and by definition p is the least integer such that $(a^k)^p = a^{kp} = e$. Then, by the previous lemma we know that $n \mid kp \implies \frac{n}{\gcd(n,k)} \mid \frac{kp}{\gcd(n,k)}$, and since $n/\gcd(n,k)$ and $k/\gcd(n,k)$ share no common factors now, then $\frac{n}{\gcd(n,k)} \mid p$. Therefore, what we have is $(a^k)^p = e \iff \frac{n}{\gcd(n,k)} \mid p$, and once again by the previous lemma, the order of a^k is $\mathcal{O}(a^k) = n/\gcd(n,k)$. □

Definition 1.11 (Generating Set). If we have some elements of a group G whose powers or inverses one can compose to form all the elements of the group, then the set of those elements is called a generating set. If the set S generates G, we write $G = \langle S \rangle$.

Example 1.12. The set of integers is completely generated by $\{1\}$, since we can add it to itself to produce all the positive integers, and its inverse -1 for all the negative integers. The same is true for $\mathbb{Z}_n$ being generated by $\{1\}$ and $\mathbb{C}_n$ being generated by $\{e^{2i\pi/n}\}$. In fact, $\mathbb{Z}_n$ and $\mathbb{C}_n$ have a very important relation between that we shall explore later.

Definition 1.13 (Cyclic Group). A group G is said to be cyclic, if $\exists\, a \in G$ such that $G = \langle \{a\} \rangle$, i.e. the entire group is generated by a single element. If $|G| = n$, then

$$G = \left\{e, a, a^2, \dots a^{n-1}\right\}$$

If G is of infinite order, then:

$$G = \left\{e, a^{\pm 1}, a^{\pm 2}, \dots\right\}$$

Proposition 1.14. *Let $G = \langle a \rangle$ be a cyclic group, then:*

1. *Every subgroup of G is cyclic.*
2. *If G is of infinite order, then every subgroup of G is of the form $H = \left\langle a^k \right\rangle$ where $k \in \mathbb{Z} - \{0\}$.*
3. *If $\mathcal{O}(a) = n$, then every subgroup of G is of the form $H = \left\langle a^{n/p} \right\rangle$ where $p \mid n$.*

Proof.

1. Given that $G = \langle a \rangle$, let H be a proper subgroup of G. Now, let m be the least integer such that $a^m \in H$, then by closure and existence of inverse, any element of the form $(a^m)^k = a^{km} \in H$, where $k \in \mathbb{Z} - \{0\}$. Now, if we assume $a^{km+1} \in H$, then $(a^{km+1})^p = a^{pkm}a^p \in H \implies a^p \in H$ for any $p \in \mathbb{Z} - \{0\}$. But then it would cease to be a proper subgroup of G since $a^p \in H \implies H = G$. Therefore only elements of the form $(a^m)^k$ are in the the group, hence $H = \langle a^m \rangle$ is a cyclic group.

   □

2. It follows as a corollary of the above proof that every subgroup of G can be written as $H = \left\langle a^k \right\rangle$ where $k \in \mathbb{Z} - \{0\}$.

   □

3. Given that $\mathcal{O}(a) = n$ then let k be least integer such that $a^k \in H$ and $\left\langle a^k \right\rangle = H$. Now, using the division algorithm we write $n = pk + r \implies e = a^n = a^{qk}a^r$, then $a^r = a^{-pk} \in H$. We chose k to be smallest integer such that a^k generates H, but $r < k$, so the only way $a^r \in h$ is for $r = 0$, so we then get $k = n/p$ and therefore every subset of G is of the form $\left\langle a^{n/p} \right\rangle$, where $p \mid n$.

   □

2 Cosets

Definition 2.1 (Coset). Ley G be a group and $H < G$ is a subgroup of G. If we have a fixed element $a \in G$, then the *left coset* of H in G is defined as:

$$aH = \{ah \mid h \in H\} \tag{5}$$

and the *right coset* of H in G is defined to be:

$$Ha = \{ha \mid h \in H\} \tag{6}$$

The cosets of H in G are equivalence classes for the relation $a \sim b \iff b = ah$ for some $h \in H$. Cosets may seem like a very specific defintion that does not seem particularly motivated, but they happen to one of the most useful tools in group theory.

Lemma 2.2. *Let G be a group and $H \leq G$. Then for $x, y \in G$, the following are equivalent:*

1. $xH = yH$.
2. $y^{-1}x \in H$.
3. $y^{-1}xH = H$.

Proof.
($1 \implies 2$) Given $xH = yH$ then $\{xh \mid h \in H\} = \{yh \mid h \in H\}$, which means that for some $h_1, h_2 \in H$, we have $xh_1 = yh_2 \implies y^{-1}x = h_2h_1^{-1} \in H$.

($2 \implies 1$) Given $y^{-1}x \in H$, then $\forall\, h_1 \;\exists\; h_2 \in H$ such that $y^{-1}xh_1 = h_2 \implies xh_1 = yh_2 \implies xH = yH$.

($1 \implies 2 \implies 3$) Given $xH = yH$ then $y^{-1}x \in H$. Now consider $y^{-1}xH = \left\{y^{-1}xh \mid h \in H\right\}$, but $y^{-1}x \in H$ so $y^{-1}xh \in H$ and thus, by closure of groups, $\left\{y^{-1}xh \mid h \in H\right\} = H$.

($3 \implies 2 \implies 1$) Given $y^{-1}xH = H$, then $y^{-1}x$ must be in H otherwise group closure of H is broken. Since $y^{-1}x \in H$, we know $xH = yH$.

□